

NP_Au SYNTHÈSE DE BRUST

(en milieu organique)

- Germes Brust 2 nm CGBP-102
- Croissance des Brust 3 nm CGBP-102A
- Croissance des Brust 4 nm CGBP-102B
- Croissance des Brust 5 nm CGBP-102C

Precise size control of hydrophobic gold nanoparticles in the 2–5 nm range

Goldmann et al. Chem Comm (2021)

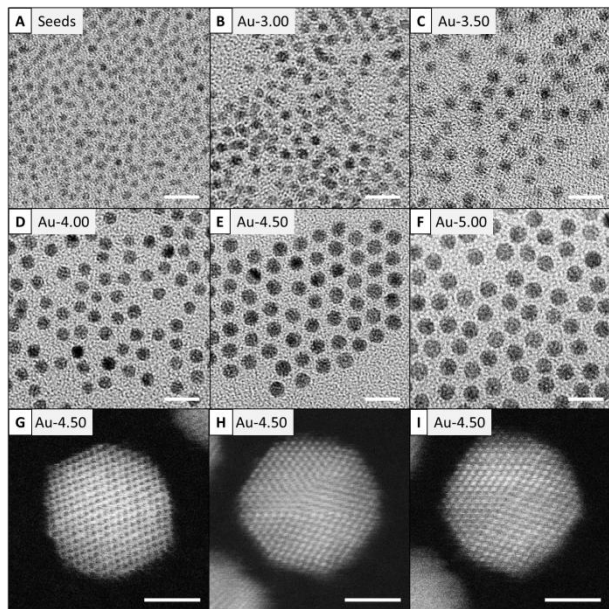


Figure 1. TEM pictures of the seeds (A) and of the nanoparticles obtained from the growth of the seeds (B-F). Scale bar: 10 nm. HRTEM pictures of the batch Au-4.50, showing monocrystalline (G) and twinned (H, I) nanoparticles. Scale bar: 2 nm.

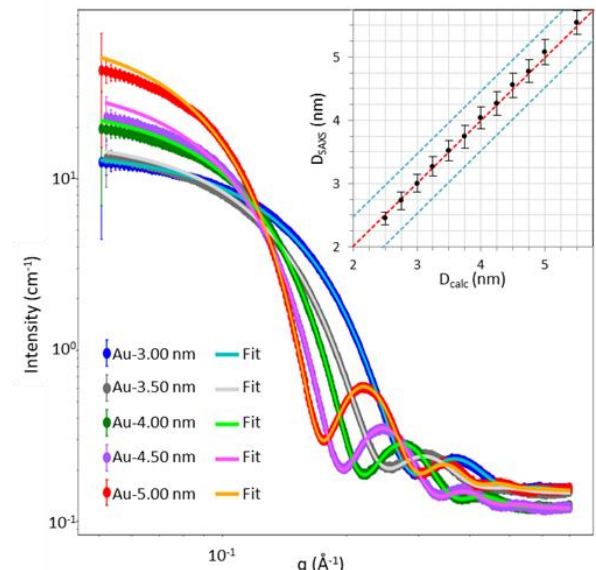


Figure 3. Small angle X-ray scattering diagrams of the nanoparticles along with their fitting curves. Inset: correlation between the expected size (D_{calc}) and the obtained size measured by SAXS (D_{SAXS}). The error bars correspond to the polydispersity σ . The red dashed line corresponds to the ideal case where $D_{\text{SAXS}} = D_{\text{calc}}$.

The blue dashed lines correspond to $D_{\text{SAXS}} = D_{\text{calc}} \pm 1$ gold monolayer.

PD : 7-10 %

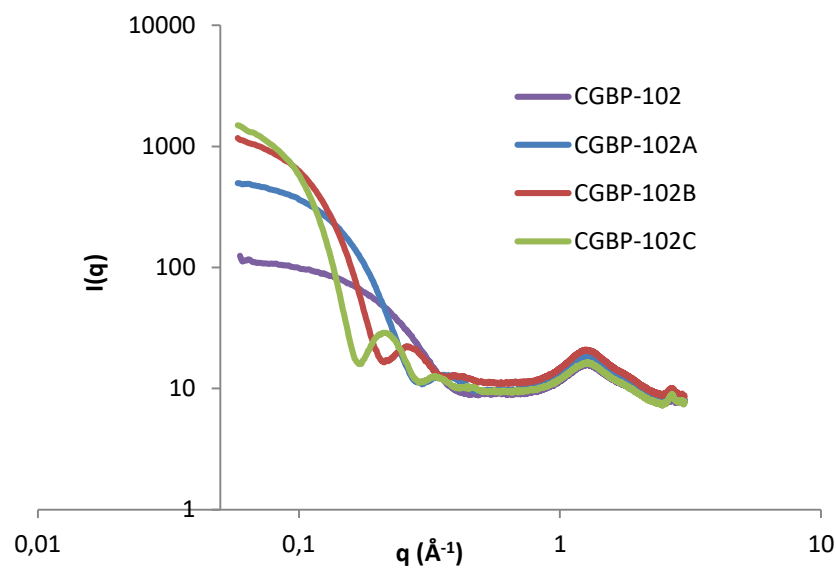
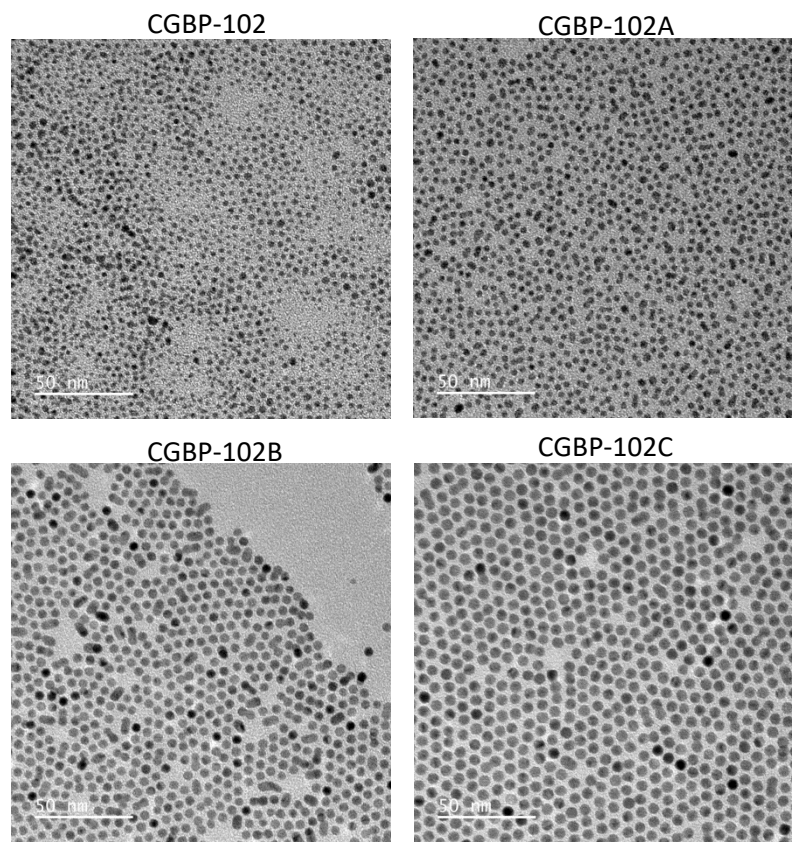
< Determining the exact morphology of the nanoparticles in a given batch is not an easy task as mixtures of shapes in various amounts can be encountered.

For such small nano-Particles like the seeds, the icosahedral morphology is expected to be more frequent than other ones like the penta-twin morphology, but this fact may be strongly influenced by the synthesis pathway >

Set n°1 Série CGBP-102 Synthèse de Brust puis croissance

- Set n°1** : NPs en milieu organique

 - Germes Brust 2 nm CGBP-102
 - Croissance des Brust 3 nm CGBP-102A
 - Croissance des Brust 4 nm CGBP-102B
 - Croissance des Brust 5 nm CGBP-102C



CGBP-102 : D = 1,98 nm, PD = 11,2%
CGBP-102A : D = 3,05 nm, PD = 9,3%
CGBP-102B : D = 4,19 nm, PD = 9,7%
CGBP-102C : D = 5,26 nm, PD = 7,7%

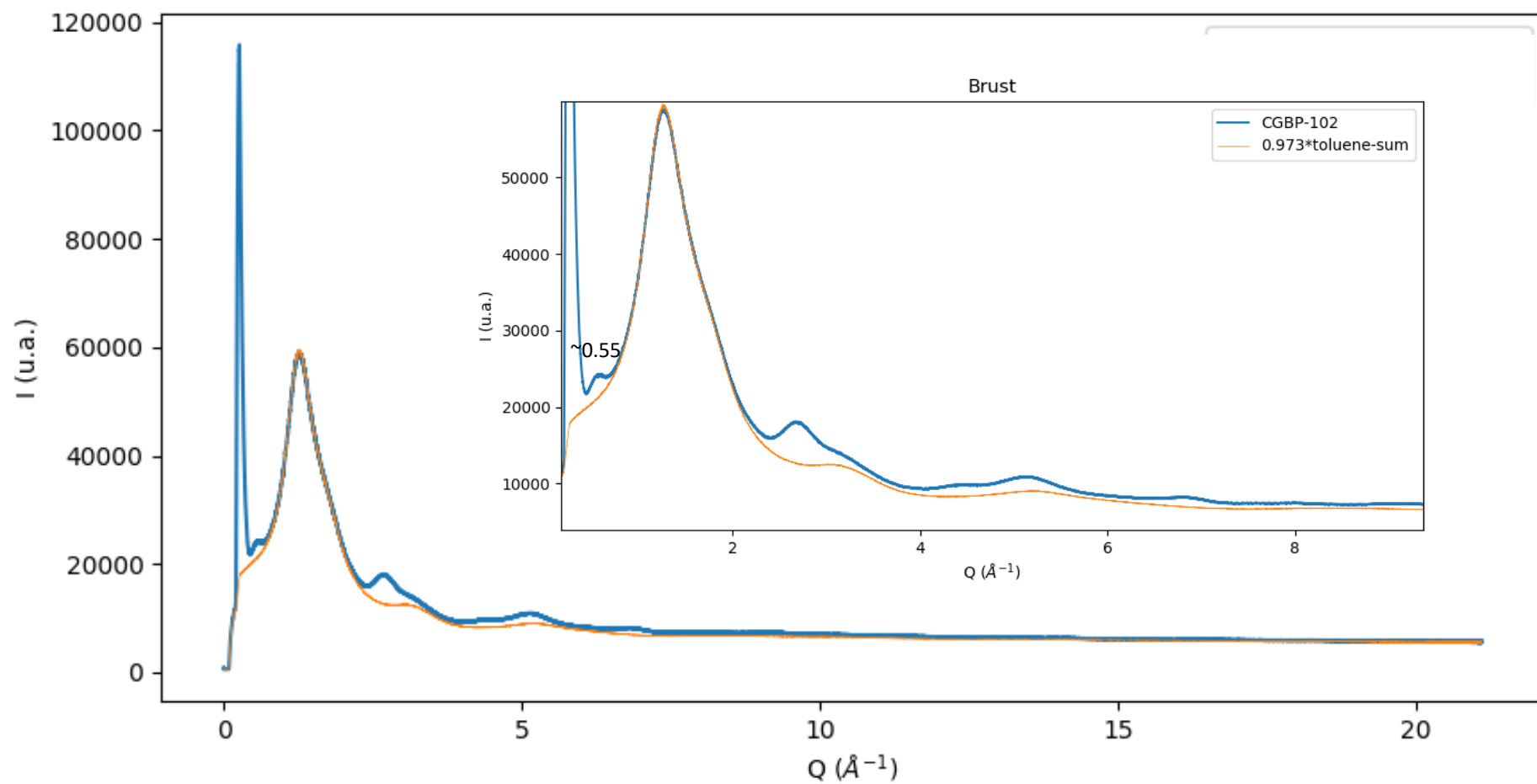
Expérience CRISTAL (SOLEIL)
NPs Au

8-10 octobre 2021

$$\lambda = 0.51391 \text{ \AA}$$

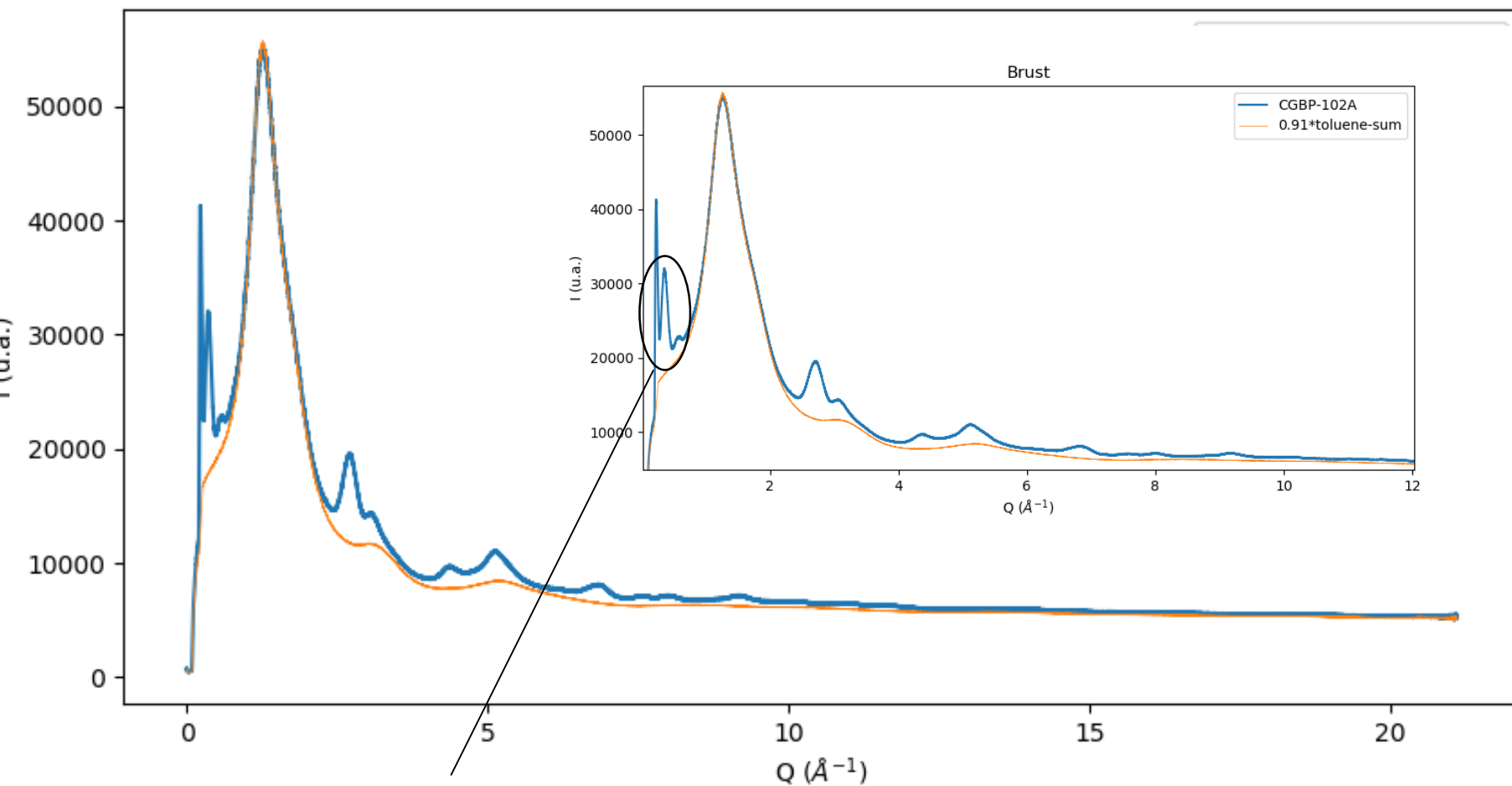
CGBP-102

Brust



CGBP-102A

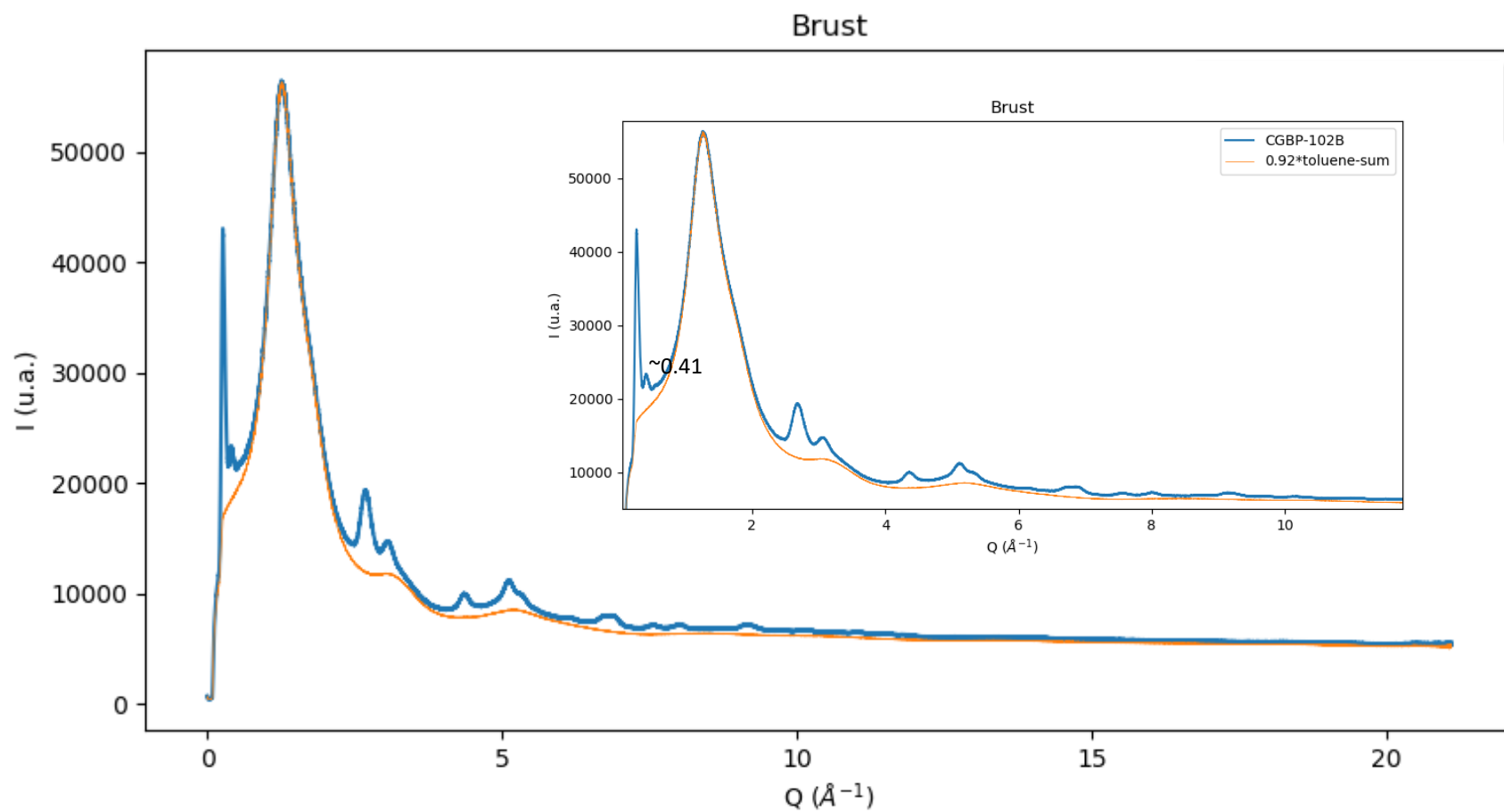
Brust



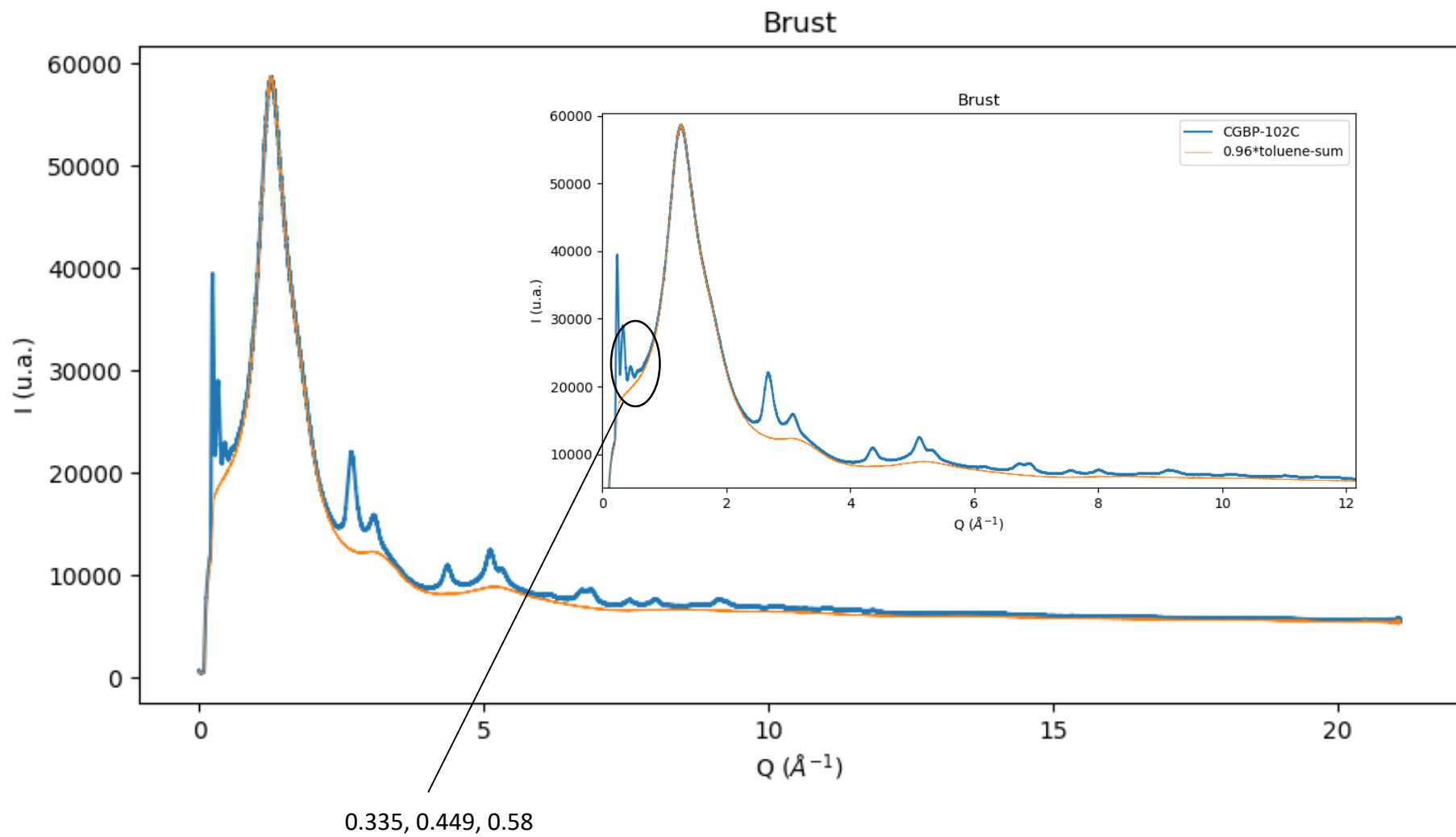
0.355, 0.449, 0.58

-> facteur de forme NPs (cf. doc Brigitte)

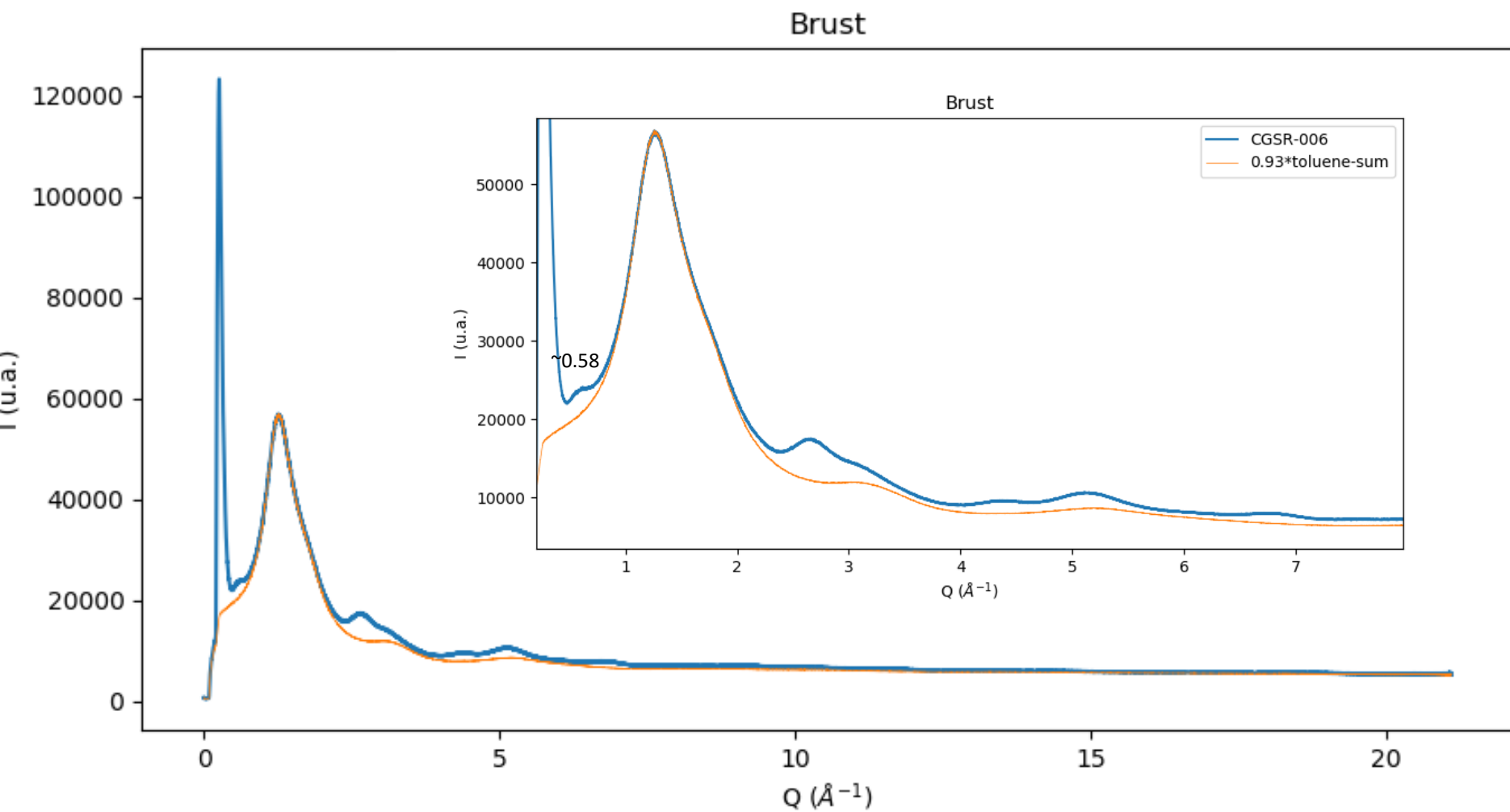
CGBP-102B



CGBP-102C



CGSR-006



Synthèse de Brust

Set n°1 : NPs en milieu organique

- Germes Brust 2 nm CGBP-102
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- Croissance des Brust 4 nm CGBP-102B
- Croissance des Brust 5 nm CGBP-102C

(Voir doc ResumeCG.pdf)

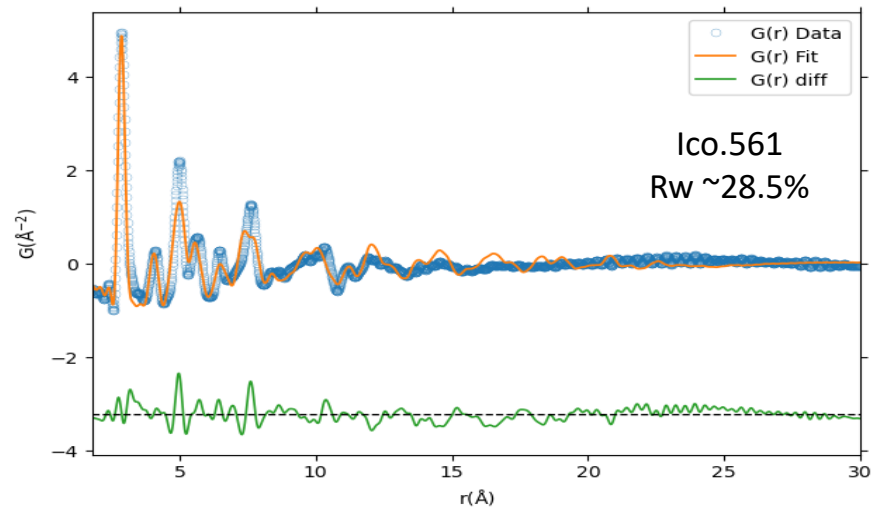
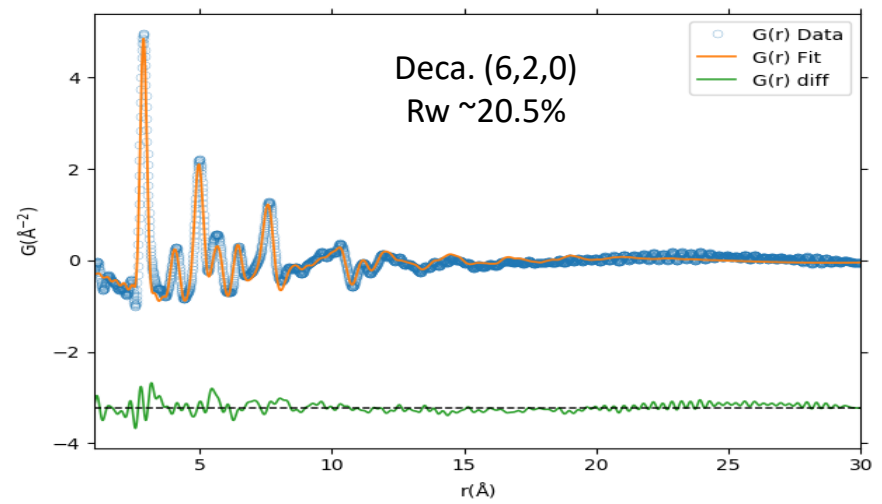
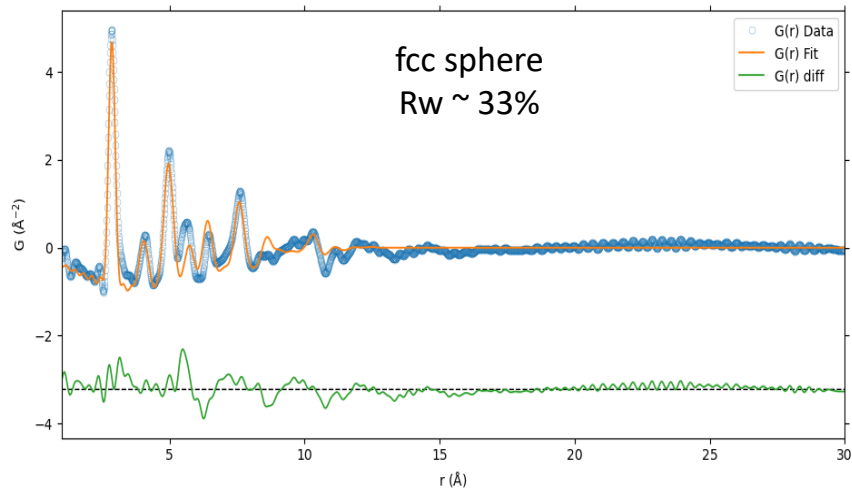
Diamètre déterminé par fit SAXS (modèle sphérique)

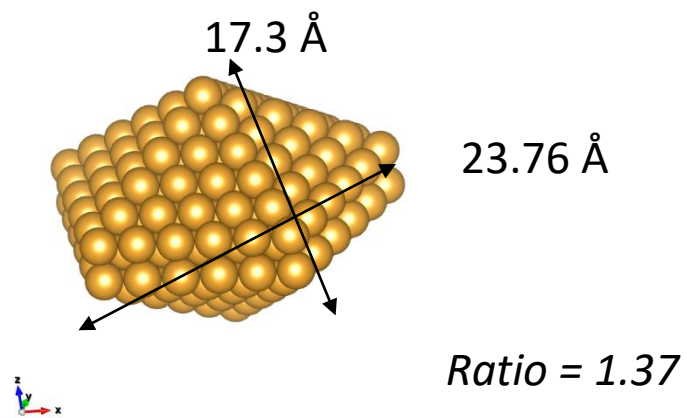
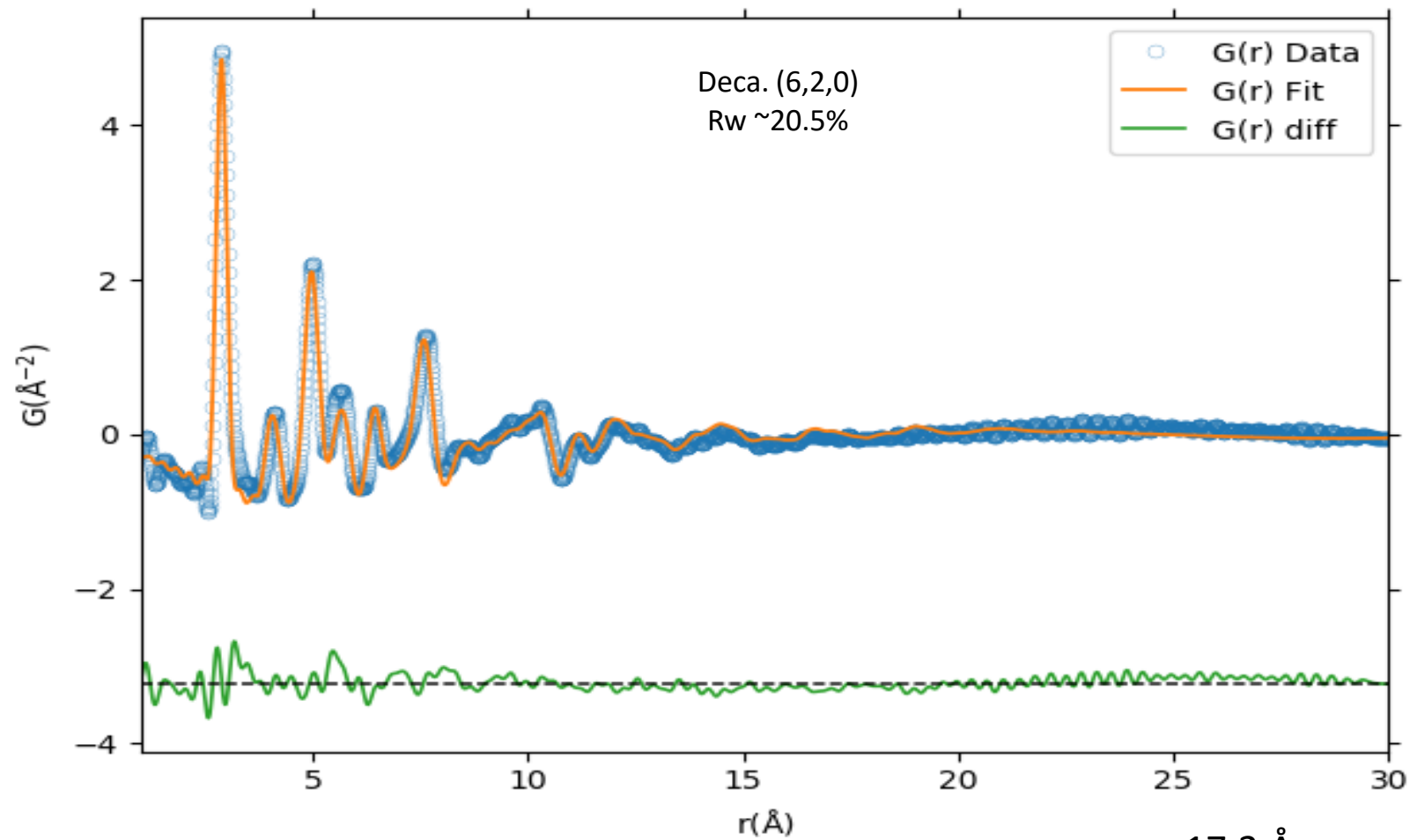
	Modix (CG)	// Cristal (BP) (Schulz model)
CGBP-102 :	19.5 Å (PD 12.3%)	N/A
CGBP-102A :	30.3 Å (PD 9.5%)	31.8 Å (PD 4.5%)
CGBP-102B :	41.18 Å (PD 11.2%)	43.2 Å (PD 6%)
CGBP-102C :	47.26 Å (PD 17.3%)	54.4 Å (PD 3%)

PDF Analysis

CGBP_102

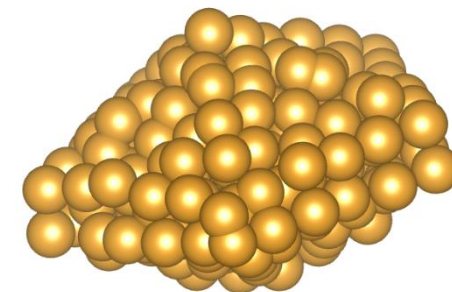
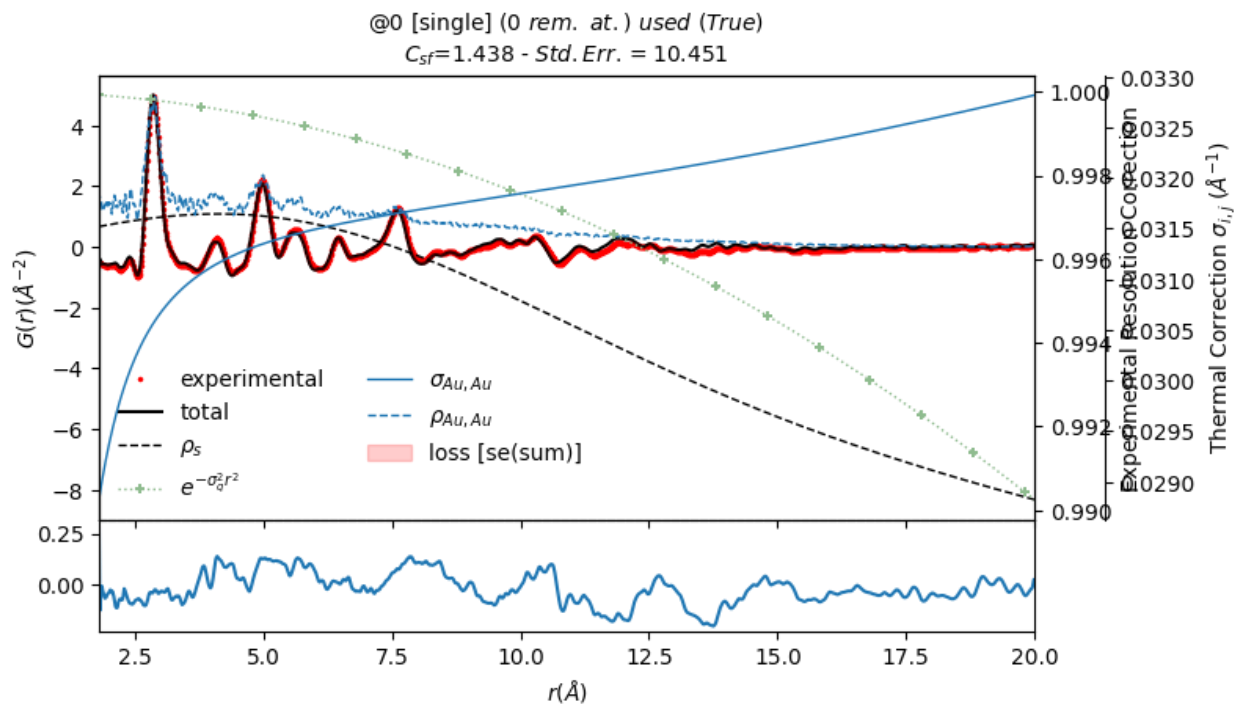
Essai avec plusieurs morphologies

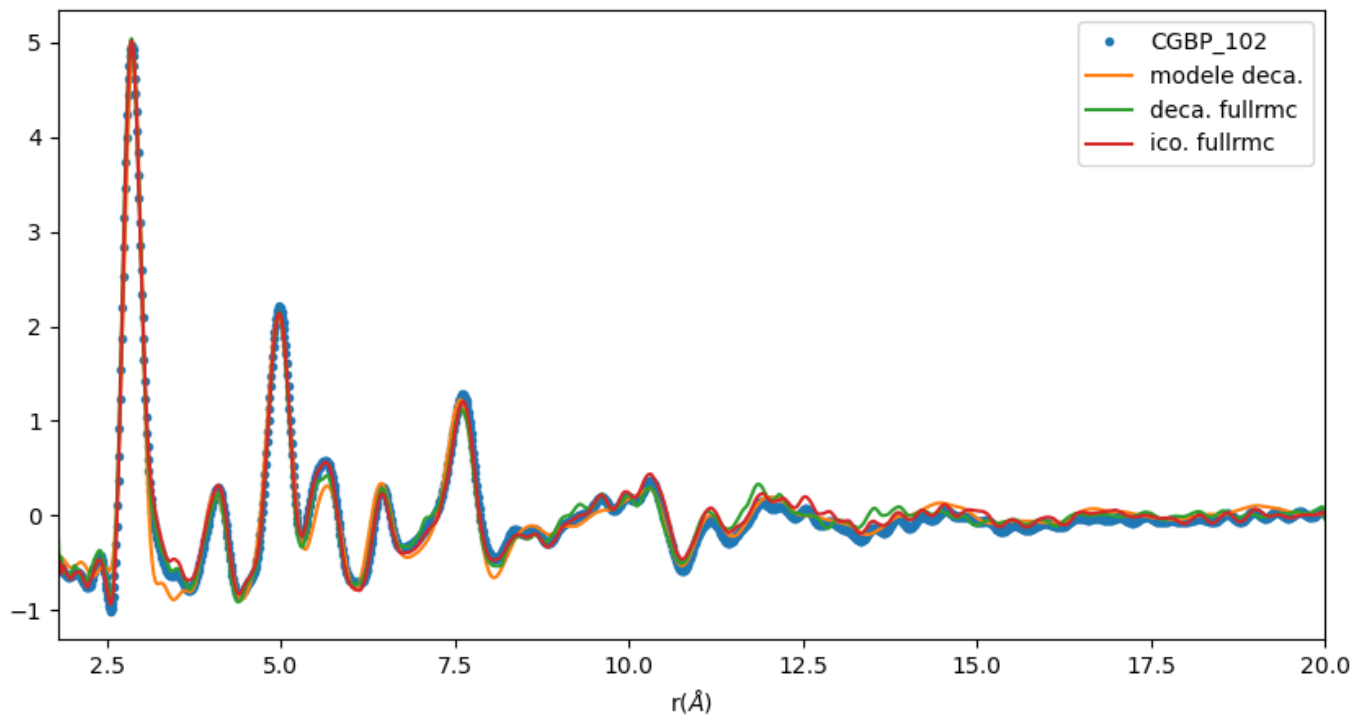




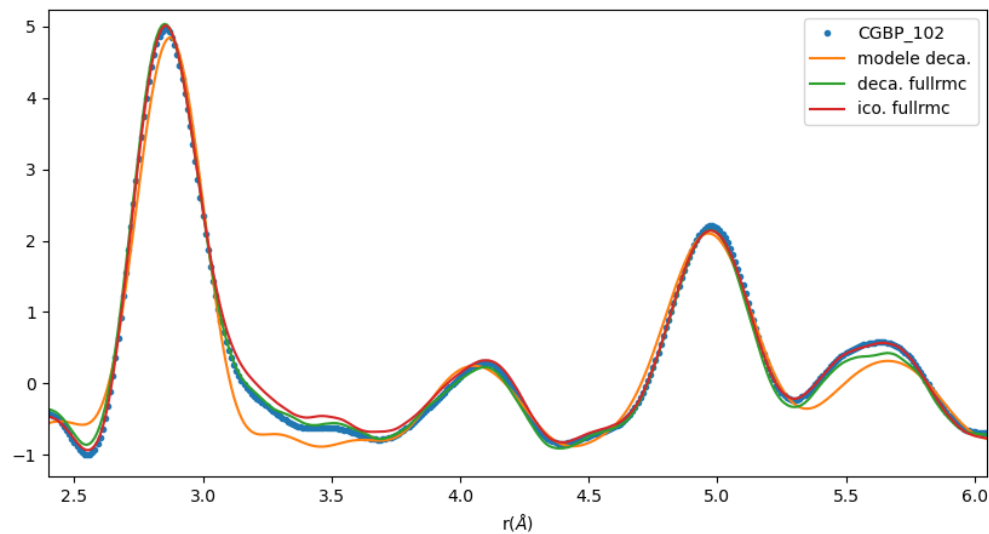
CGBP_102

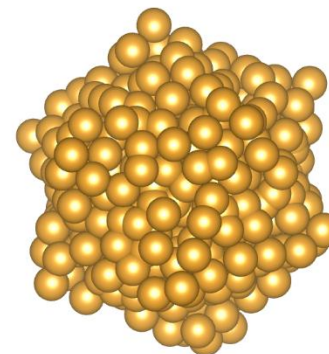
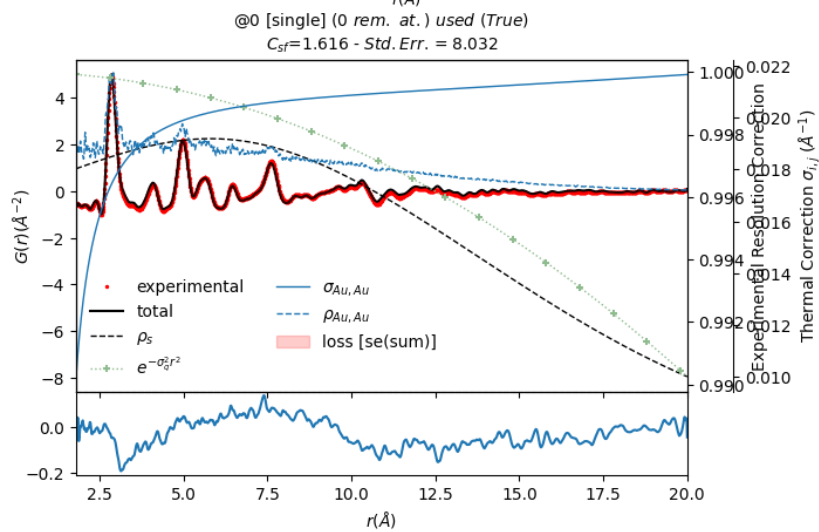
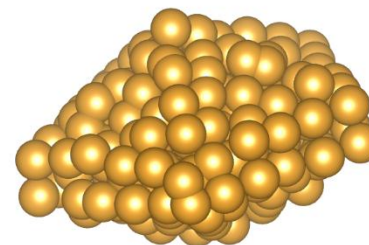
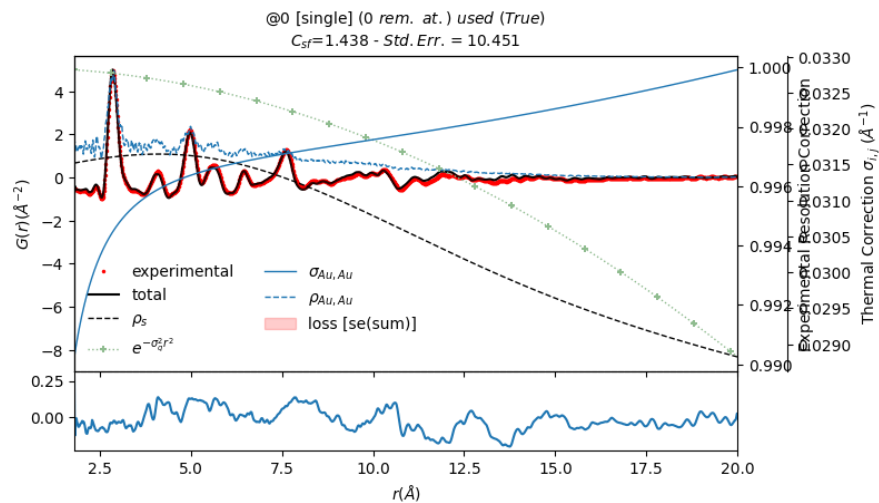
Affinement RMC (Reverse Monte Carlo) method





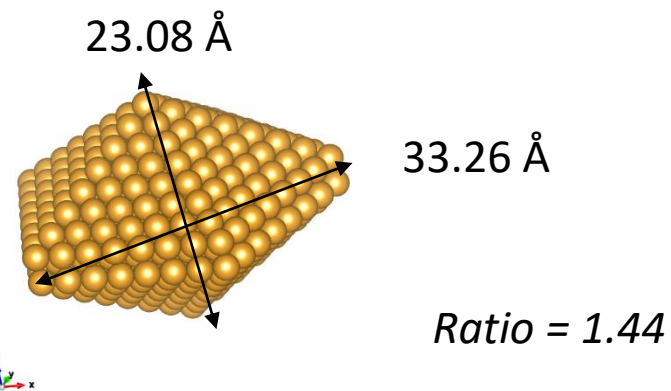
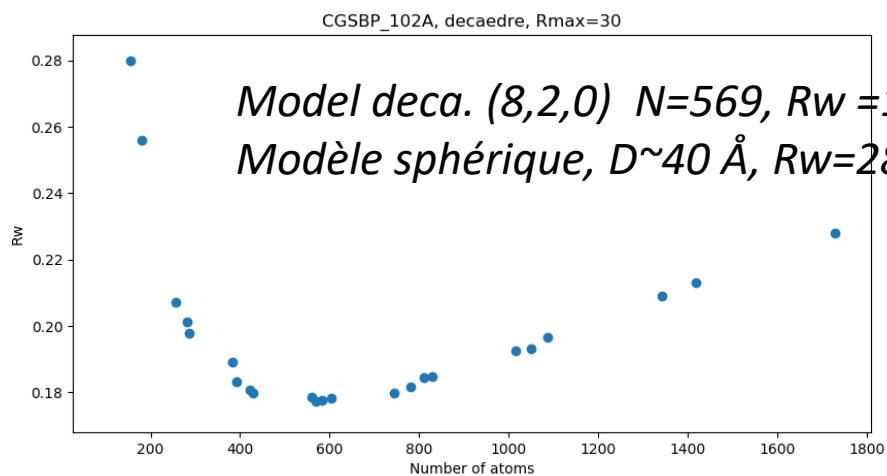
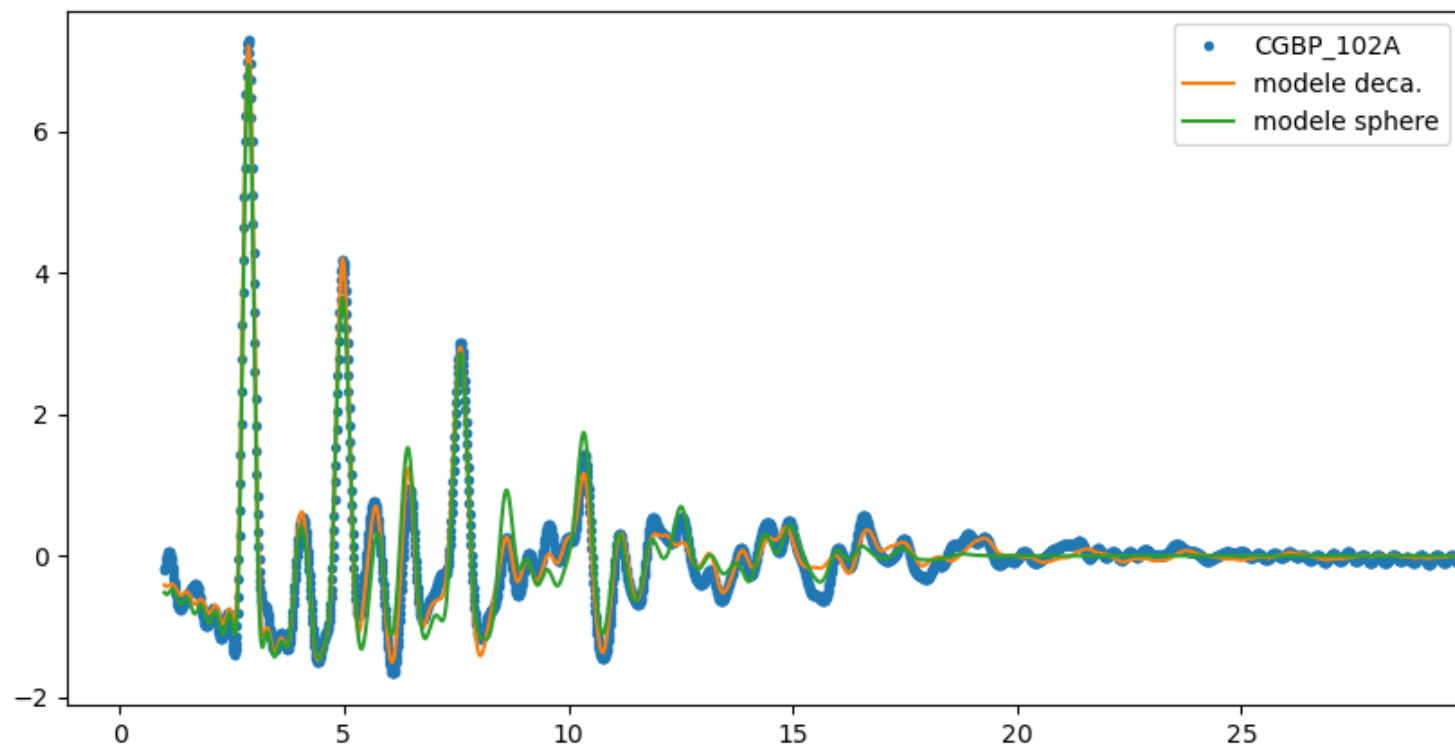
Ico. légèrement mieux





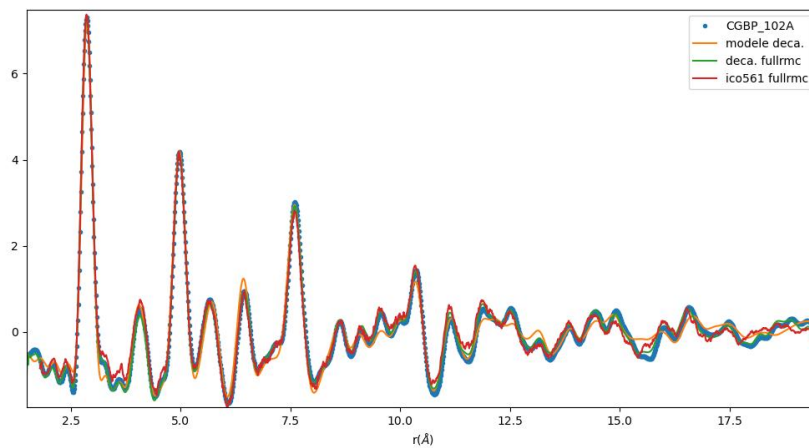
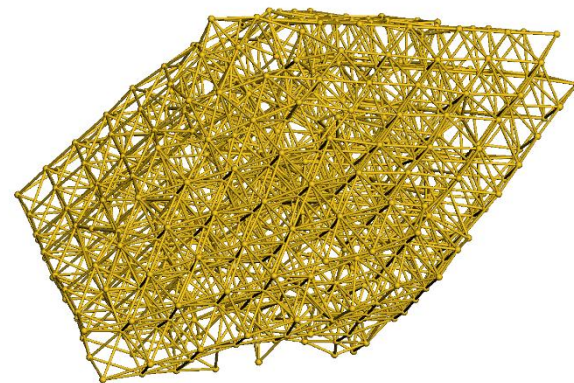
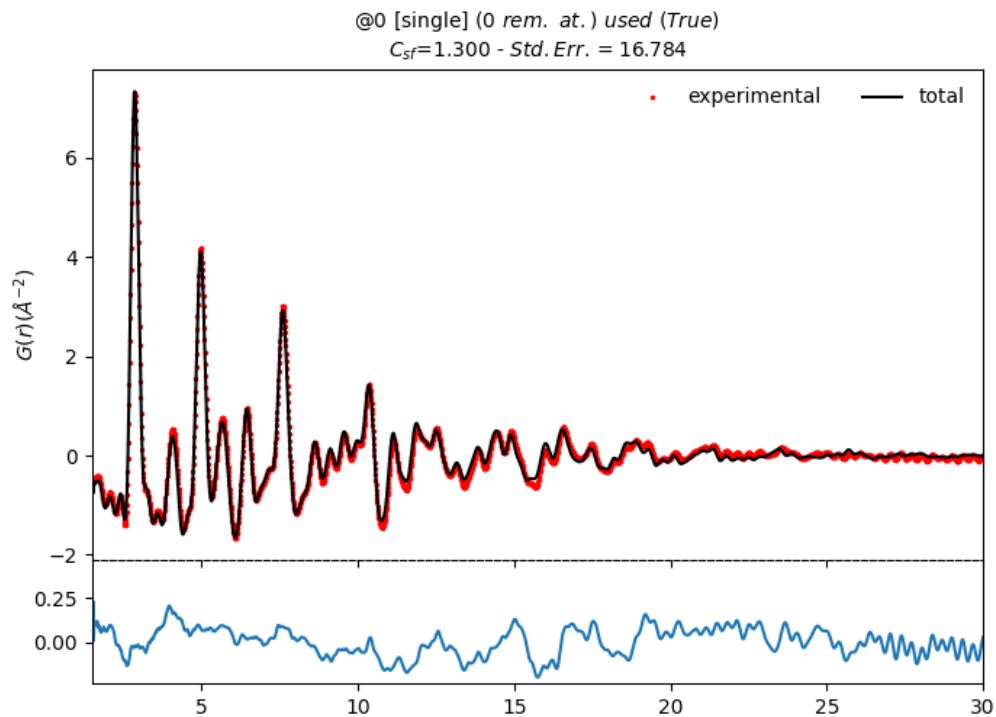
CGBP_102A

CGBP_102A



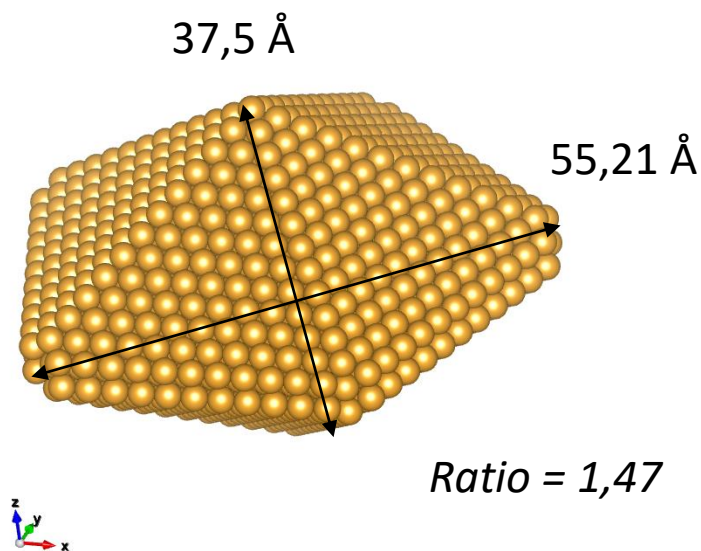
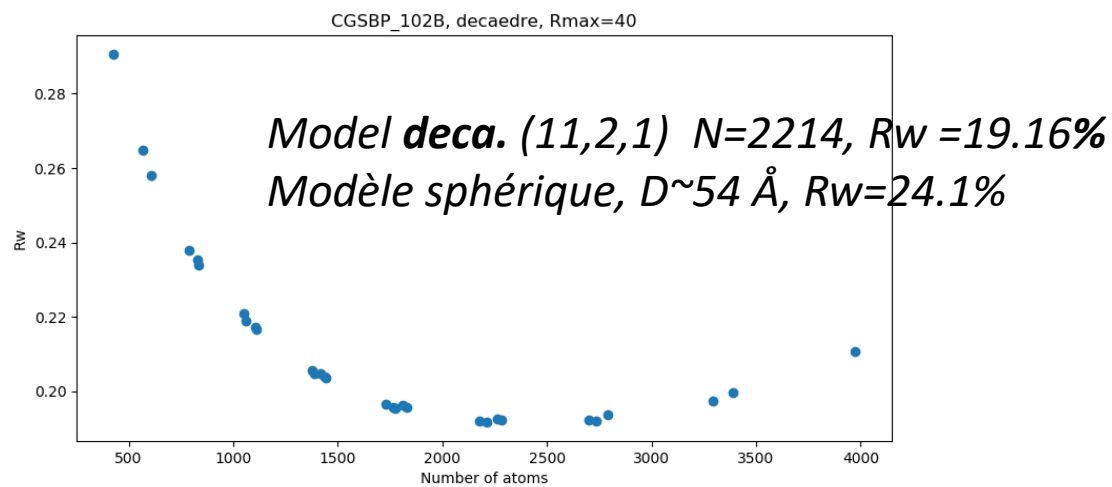
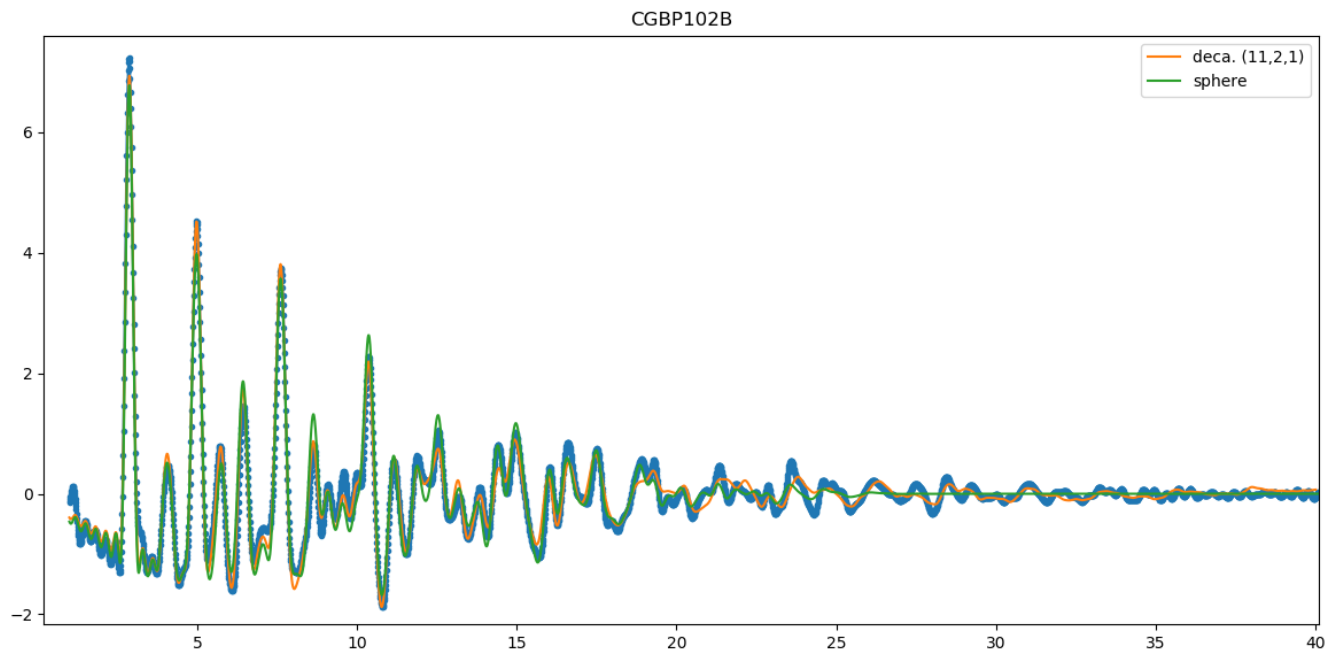
CGBP_102A

Affinement RMC method



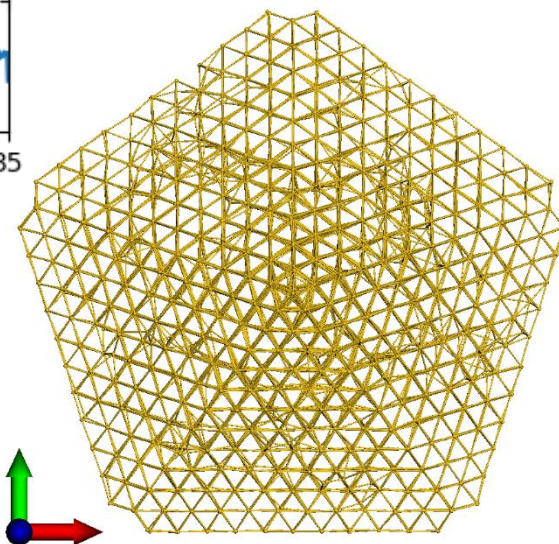
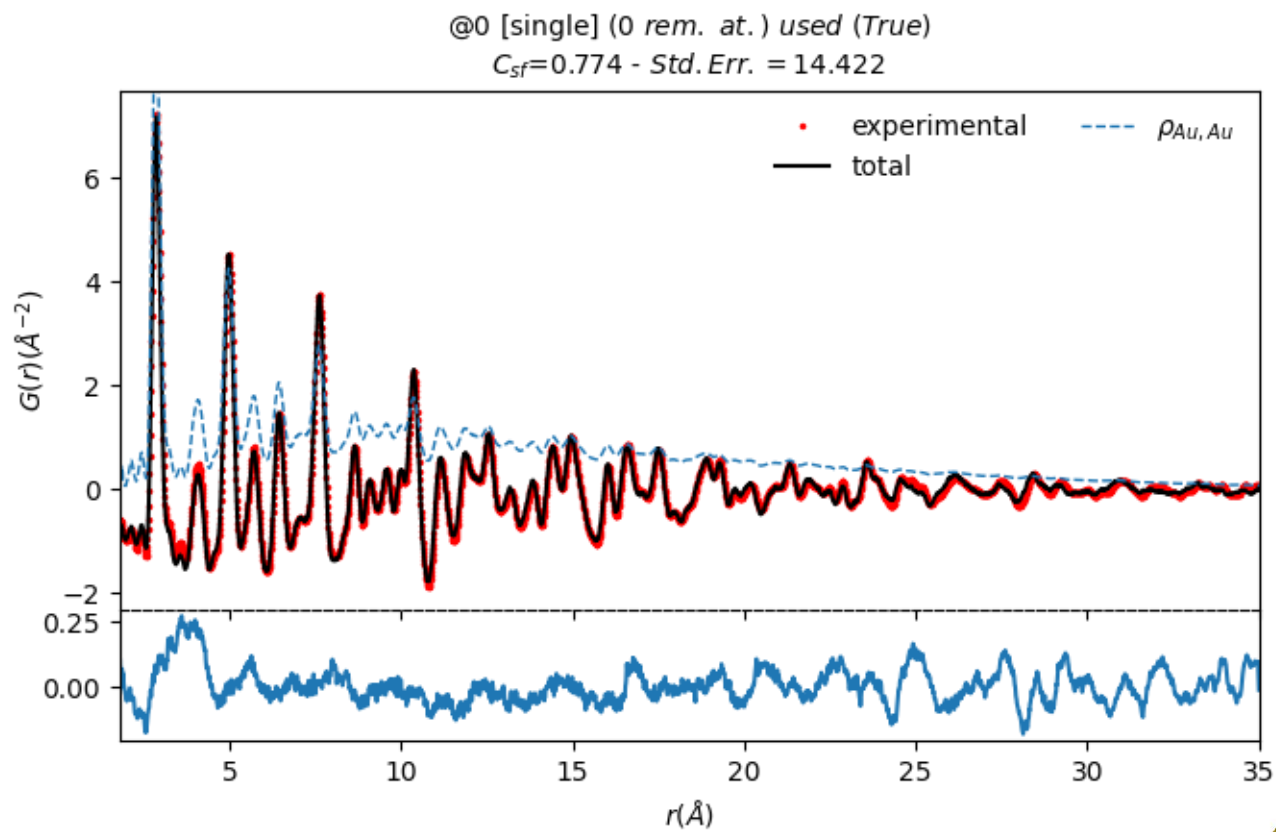
CGBP_102B

CGBP_102B



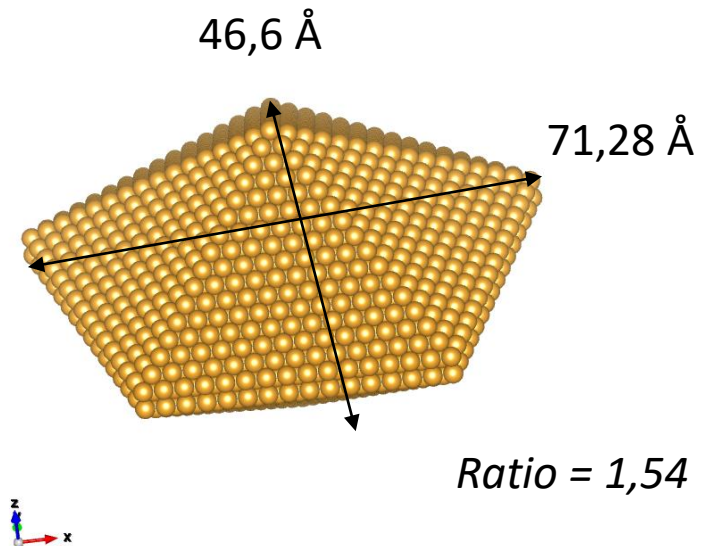
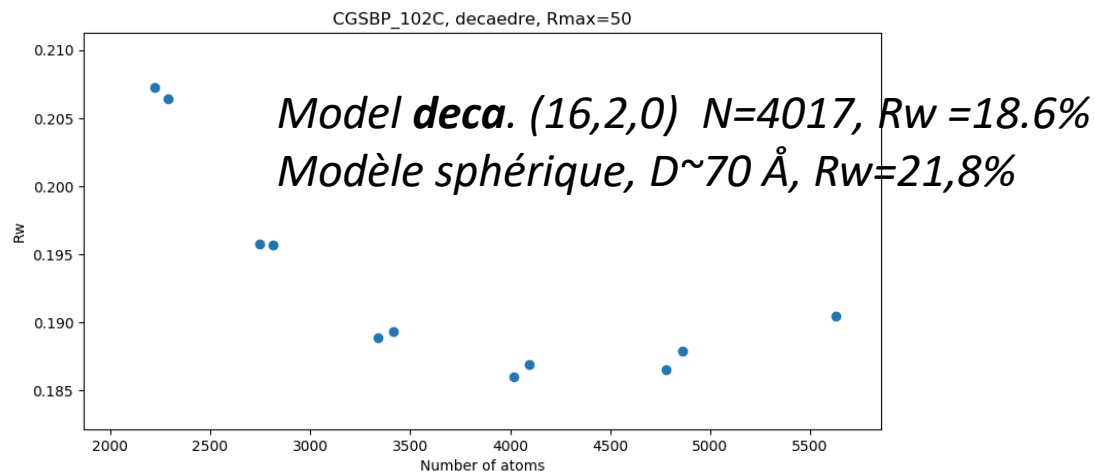
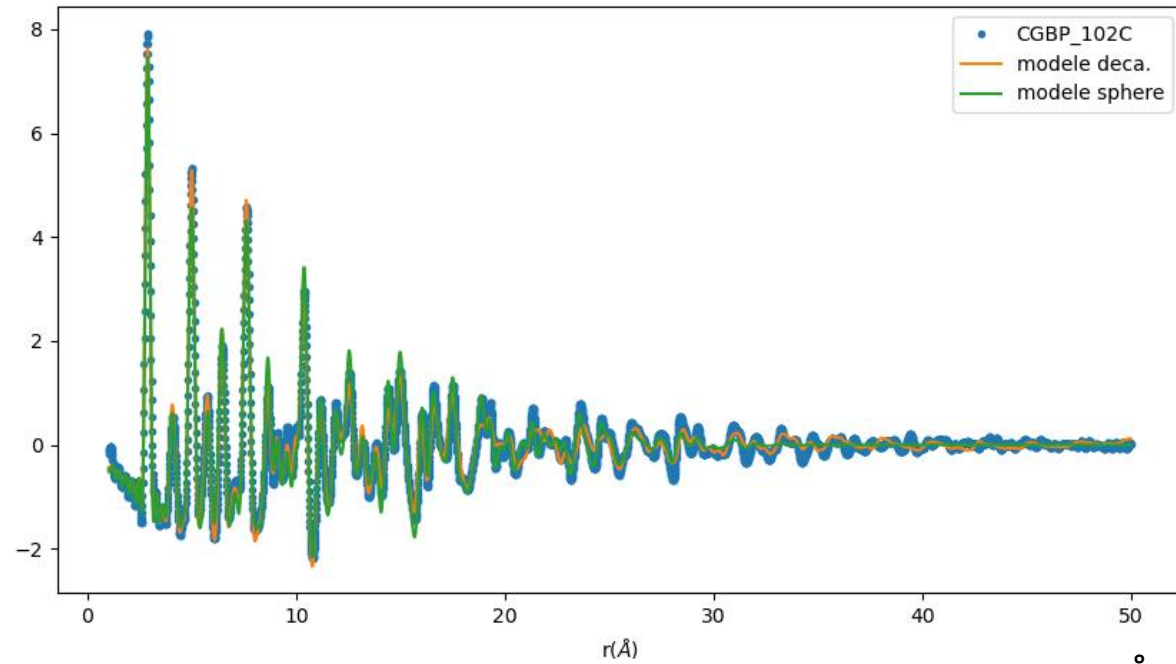
CGBP_102B

Affinement RMC method



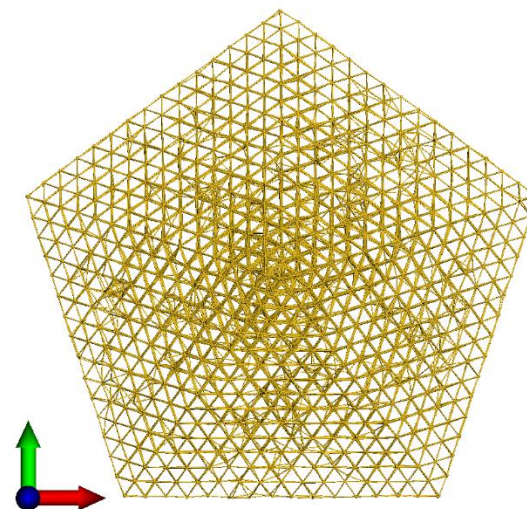
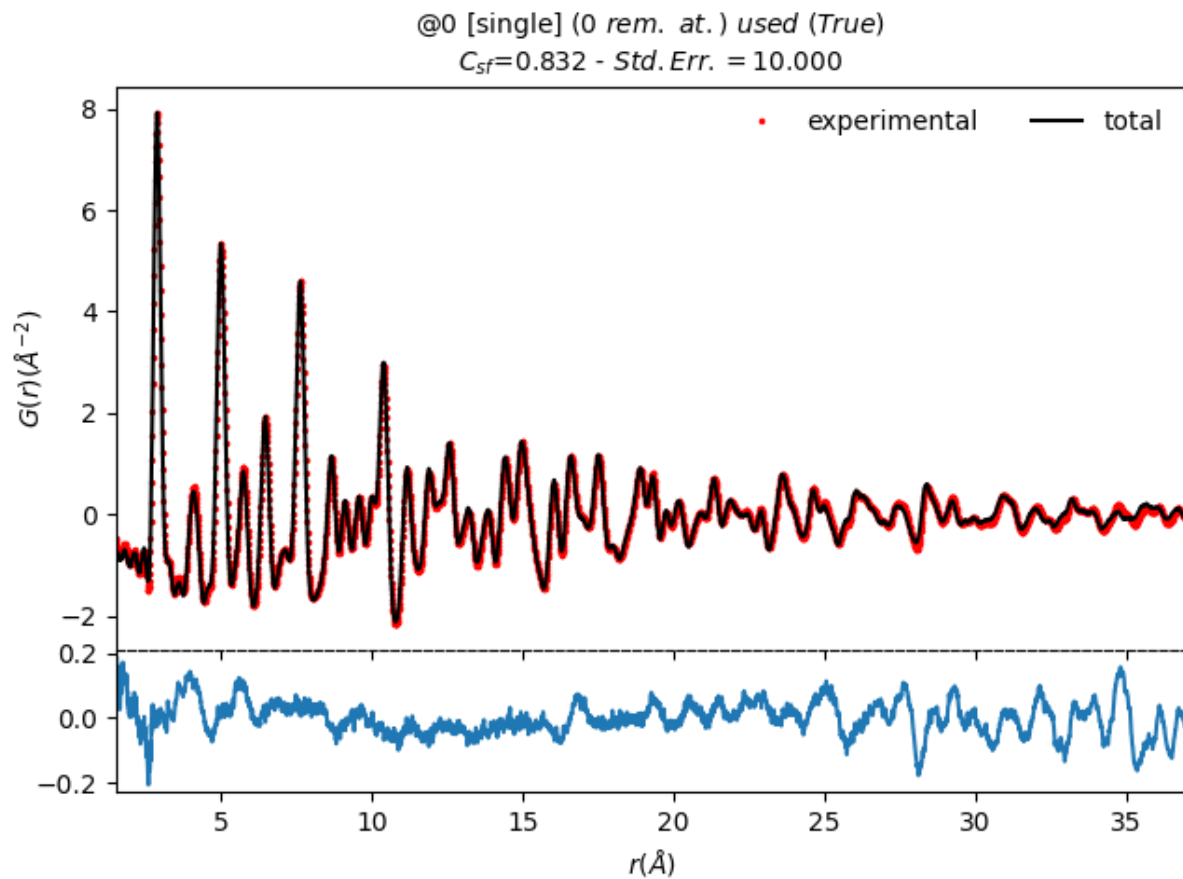
CGBP_102C

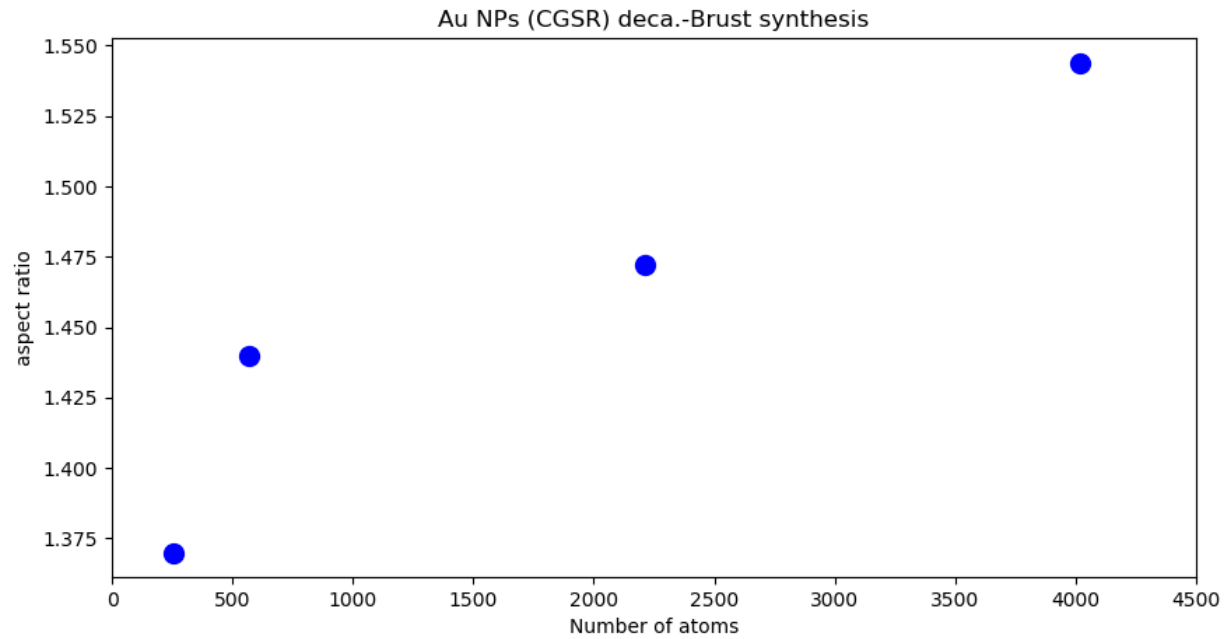
CGBP_102C



CGBP_102C

Affinement RMC method





- NP deca.

- Rapport d'aspect augmente avec la taille de la NP

`ase.cluster.Decahedron(symbol, p, q, r, latticeconstant=None)` [\[source\]](#)

Return a decahedral cluster.

Parameters:

- **symbol** (*Chemical symbol (or atomic number) of the element.*)
- **p** (*Number of atoms on the (100) facets perpendicular to the five*)
- **axis.** (*fold*)
- **q** (*Number of atoms on the (100) facets parallel to the five fold*)
- **facets.** (*axis. q = 1 corresponds to no visible (100)*)
- **r** (*Depth of the Marks re-entrence at the pentagon corners. r = 0*)
- **re-entrence.** (*corresponds to no*)
- **(optional)** (*latticeconstant*)
- **ase.data.** (*then it is extracted form*)